Stochastic collapse of spatial cooperation: Perturbation geometry and resource alignment

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The fate of a finite disturbance in a spatial system can depend on both its initial geometry and the environment through which it propagates. Here we investigate this dependence in a stochastic spatial evolutionary game coupled to a renewable, diffusing resource. Compact and dispersed perturbations containing the same number of defectors and the same resource deficit exhibit markedly different finite-time outcomes: dispersed perturbations are more likely to establish growth. Exact first-visit relations link this contrast to interfacial activity and the directional bias of strategy updates. As disturbances expand, they develop resource-depleted interiors surrounded by interfacial resource layers. Translating the resource field preserves the strategy configuration, resource distribution, and resource autocorrelation, yet changes the probability of subsequent collapse. Short and long translations produce responses of opposite sign. Exact resource-balance and sensitivity relations connect these effects to strategy–resource covariance and local payoff differences. Redistributing resources along a fixed interface can also reverse the initial growth response, although this ordering need not persist during subsequent evolution. These findings demonstrate that relative strategy–resource alignment contains information about finite-disturbance robustness that is not captured by the statistics of either field separately.

I. INTRODUCTION

Finite-amplitude perturbations probe a form of stability that is not captured by the response to infinitesimal deviations. Even when a state is locally stable, a sufficiently large disturbance may redirect its evolution toward a different macroscopic outcome. This distinction is central to ecological resilience [1, 2] and to basin-based measures of nonlinear stability [3]. In spatially extended systems, perturbation magnitude is not the only relevant quantity: the initial arrangement of disturbed sites also determines where interfaces form and how they can propagate through the surrounding medium.

Spatial organization connects local interactions to collective dynamics [4, 5]. Pattern formation and front propagation provide useful descriptions of how interfaces emerge, move, and reorganize extended structures [6, 7]. In ecological systems, feedback between populations and transported resources can reshape spatial patterns and alter the response to perturbations [8–10]. Spatial evolutionary games offer a complementary microscopic setting in which the competing domains correspond to different strategies. Local imitation and interaction structure generate correlations and persistent spatial patterns [11, 12], although spatial organization does not universally favor cooperation [13]. Results on graph-structured populations and network reciprocity further show how interaction topology affects selection [14–16], connecting spatial game dynamics to nonequilibrium statistical physics [17–20].

Environmental feedback adds a second evolving field to this picture [21]. Ecological public-goods models have shown that dispersal and population feedback can generate spatial structure [22]. Game–environment feedback can also drive depletion, recovery, and oscillatory dynamics, with outcomes depending on heterogeneity and relative time scales [23–25]. In stochastic games and games with changing interaction conditions, strategies modify the environment in which subsequent competition takes place [26, 27]. When these effects are local, strategy and resource distributions can form coupled spatial patterns [28–30]. The resource field is therefore not merely a background parameter: it records and responds to the spatial history of the competing populations.

This coupling raises a question that cannot be resolved from the statistics of either field alone. Can two states with the same strategy configuration and the same resource-value distribution and autocorrelation have different probabilities of collapse because the resource pattern is positioned differently relative to the strategy interface? Comparing independently evolved configurations does not isolate this effect, because interface geometry, resource abundance, and spatial correlations generally change together. Translating only the resource field on a periodic lattice provides a controlled comparison: its distribution and autocorrelation remain unchanged, as does the entire strategy configuration, while the relative alignment of the two fields changes.

Here we investigate finite disturbances in a stochastic lattice evolutionary game coupled to resource renewal, depletion, and diffusion. We first compare compact and dispersed perturbations with identical defector counts

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and initial resource loss to isolate the effect of disturbance geometry. We then translate the resource field of grown disturbances to test the role of relative alignment in subsequent collapse. Spatial measurements and first-passage probabilities connect the geometric contrast to disturbance establishment and the alignment contrast to interfacial resource structure. Exact first-visit, resource-balance, and sensitivity relations identify the microscopic terms underlying these observations. Finally, resource permutations along a fixed boundary distinguish immediate interfacial responses from their subsequent evolution. Throughout, collapse and recovery are defined by finite-time first-passage criteria [31], rather than by an inferred asymptotic phase boundary.

II. MODEL AND OBSERVABLES

We consider a periodic square lattice containing $N = L^2$ sites, each with four nearest neighbors. Each site i carries a binary strategy $s_i \in \{0, 1\}$, with $s_i = 1$ representing cooperation, and a resource level $r_i \in [0, 1]$. One Monte Carlo step (MCS) consists of a synchronous resource update followed by one strategy-update visit to each site in random order. Resource dynamics combine local renewal or depletion with nearest-neighbor diffusion:

$$\begin{aligned} u_i &= r_i + g s_i(1 - r_i) - \delta(1 - s_i)r_i, \\ r_i^+ &= (1 - D)u_i + \frac{D}{4}\sum_{j\sim i} u_j. \end{aligned} \tag{1}$$

For $0 \le g, \delta, D \le 1$, the update preserves $r_i \in [0, 1]$ in exact arithmetic; numerical clipping to this interval is therefore a safeguard rather than an additional model rule.

During the strategy-update stage, the resource field remains fixed. Sites are visited in a uniformly random order, and a visited site i samples a nearest neighbor j. The strategy of j is copied according to the Fermi rule

$$p_{i\leftarrow j} = \frac{1}{1 + \exp[(\Pi_i - \Pi_j)/K]}. \tag{2}$$

where the payoffs are evaluated using the strategy configuration at the time of the visit. Writing $k_C(i) = \sum_{j\sim i} s_j$ for the number of cooperative neighbors, the payoffs are

$$\begin{aligned} \Pi_C(i) = R(r_i^+)k_C(i), \qquad \Pi_D(i) = b\, k_C(i), \\ R(r) = \max\{0, 1 + \lambda\sigma(2r - 1)\}. \end{aligned} \tag{3}$$

The Fermi rule is standard in stochastic evolutionary games [32, 33]. We use $L = 48$, $b = 1.16$, $D = 0.4$, $g = 0.1$, $\delta = 0.5$, $\lambda = 0.5$, $\sigma = 1$, and $K = 0.1$ as reference parameters. At this coupling, $R(r) = 0.5 + r$ remains on the positive branch. All parameters are dimensionless, and time is measured in MCS. In the absence of mutation, the two homogeneous strategy configurations are absorbing under imitation. The implementation clips the Fermi exponent to $[-30, 30]$; the Supplemental Material [34], Sec. SI, distinguishes this numerical convention from the smooth diagnostic used in the analysis.

A. Resource balance and spatial correlation

Let $\rho = N^{-1}\sum_i s_i$, $\bar{r} = N^{-1}\sum_i r_i$, and $\langle sr\rangle = N^{-1}\sum_i s_i r_i$. Diffusion conserves the sum of its input field on the periodic lattice. The change of the mean resource during a complete resource substep is therefore

$$\begin{aligned} \bar{r}^+ - \bar{r} &= g\rho - \delta\bar{r} + (\delta - g)\langle sr\rangle \\ &= g\rho(1 - \bar{r}) - \delta(1 - \rho)\bar{r} \\ &\quad + (\delta - g)C_{sr}, \end{aligned} \tag{4}$$

where $C_{sr} = \langle sr\rangle - \rho\bar{r}$ is the spatial strategy–resource covariance. The first two terms in the covariance form describe renewal and depletion evaluated at the mean strategy and resource levels. The covariance term measures their spatial association and cannot generally be reconstructed from ρ and $\bar{r}$ alone. Thus the pair $(\rho, \bar{r})$ does not form a closed description of the resource dynamics. Diffusion cancels from the instantaneous total-resource balance on the periodic lattice, but it reshapes the resource field and thereby influences later resource and strategy updates.

B. Equal-budget perturbations and fate probabilities

Starting from an all-cooperator configuration with unit resources, we convert exactly $m = 16$ sites to defectors and set the resources at those sites to zero. For compact perturbations, the selected sites are the 16 closest to a common center in squared periodic Euclidean distance, with ties resolved by a fixed coordinate ordering. For dispersed perturbations, 16 sites are sampled uniformly without replacement. The two protocols therefore have identical initial defector counts and resource deficits, differing only in the spatial placement of the perturbation. We also consider a single-defector seed as a size control.

We label persistent depletion as event B when both $\rho \le 0.2$ and $\bar{r} \le 0.2$ hold for 250 consecutive MCS. Recovery (event A) is recorded when all sites cooperate and $\bar{r} \ge 0.6$, provided event B has not already occurred. Each trajectory retains its first recorded event. Trajectories satisfying neither criterion by time t belong to the unresolved class U, giving

$$P_A(t) + P_B(t) + P_U(t) = 1. \tag{5}$$

The persistence condition defines an operational first-passage event; it does not make the depleted threshold region mathematically absorbing. Unresolved trajectories remain in the denominators of all numerical event probabilities. Observation windows, simulation counts,

and uncertainty estimates are given in Sec. SII of the Supplemental Material. Conditional fate probabilities for continuations from grown configurations are defined in Eq. (16).

III. GEOMETRY-DEPENDENT COLLAPSE

Perturbations with the same initial defector count and resource deficit need not have the same probability of collapse. A dispersed perturbation creates more cooperator–defector contacts than a compact patch, altering the opportunities for early growth. Figures 1 and 2 show that this geometric distinction is most consequential near the crossover between recovery and persistent depletion. At stronger diffusion, the collapse probability changes rapidly with the defector payoff b; at weaker diffusion, a substantial fraction of trajectories remains unresolved within the observation window.

The time required for half of the initial ensemble to confirm collapse is

$$t^B_{1/2} = \inf\{t : \widehat{P}_B(t) \geq 1/2\}. \tag{6}$$

The dispersed preparation reaches a 50% collapse probability in part of the parameter plane where the compact preparation does not. A low measured collapse probability, however, does not necessarily imply recovery: the disturbance may have disappeared, or it may remain active without satisfying either stopping criterion. The unresolved category must therefore be shown explicitly when interpreting finite-time regime maps.

Figure 2 makes the displacement of the crossover more apparent. Dispersed seeds reach comparable collapse probabilities at lower values of b than compact seeds, and increasing diffusion shifts both crossovers in the same direction. Their contrast is concentrated near the crossover rather than spread uniformly across parameter space. The larger separation in collapse probabilities is also present in the longer observation window discussed below. Because unresolved trajectories are retained, the observed ordering does not depend on classifying them as either recovery or collapse. These results establish a finite-time, geometry-dependent difference in collapse risk; they do not by themselves establish an asymptotic phase transition, which would require the size and time dependence used to identify absorbing-state transitions [35, 36]. Quantitative estimates and their uncertainties are given in Sec. SII of the Supplemental Material.

IV. INTERFACIAL BIAS AND SPATIAL GROWTH

To connect initial geometry to collapse risk, we examine the motion and organization of cooperator–defector interfaces. Figure 3 compares illustrative trajectories at matched defector fractions, rather than at matched times. Compact seeds initially expand from a single patch, whereas dispersed seeds establish growth through several separated neighborhoods. Both can eventually develop a connected defector-rich region with a depleted interior and a surrounding resource layer. The distinct early morphologies motivate a quantitative description of interfacial activity and directional bias.

A. Geometry and the direction of local motion

Interface geometry determines both the number of possible strategy-changing contacts and their local payoff differences. An isolated defector is surrounded by cooperators and can receive payoff contributions from all four neighboring sites. In a compact defector cluster, interior defectors have fewer cooperative neighbors, while the shorter boundary offers fewer opportunities for imitation across strategies. The microscopic organization of an interface is known to matter for spatial game dynamics [37]; here its contribution can be expressed directly. All quantities are evaluated on the state after the resource update and before the first strategy visit. For an undirected interfacial edge $e = (u = C, v = D)$, let $x_e = [\Pi_D(v) - \Pi_C(u)]/K$. We define the signed interfacial bias as

$$J = \frac{1}{4}\sum_{e\in\mathcal{I}} \tanh(x_e/2). \tag{7}$$

Under the smooth imitation rule, uniform sampling of a site and one neighbor during the first strategy visit gives

$$\begin{aligned} \mathbb{E}[Z \mid s, r] &= J/N, \\ \mathbb{E}[Z^2 \mid s, r] &= P/(4N), \\ \mathrm{Var}(Z \mid s, r) &= P/(4N) - (J/N)^2, \end{aligned} \tag{8}$$

Here $P = |\mathcal{I}|$ is the number of undirected interfacial edges and Z is the change in defector count during the first strategy visit, conditional on the fixed pre-visit state. Equation (8) separates activity from direction: $P/(4N)$ is the probability of a population-changing event at that visit, whereas J/N is its signed mean. A long interface can therefore remain active even when opposing update biases nearly cancel.

For a state sampled before its resource update, $J_{\mathrm{pre}}(s, r) = J(s, T_s(r))$ includes that update once. An isolated defector has four cooperative neighbors, each with three cooperative neighbors of its own. On the positive reward branch its initial bias is

$$J_{\mathrm{iso}} = \tanh\left[\frac{4b - 3\{1 + \lambda\sigma(1 - D/2)\}}{2K}\right] \tag{9}$$

The isolated-defector expression illustrates how payoff advantage and diffusion act together at the initial boundary. At the reference parameters, the ensemble-averaged initial bias is positive for dispersed perturbations and

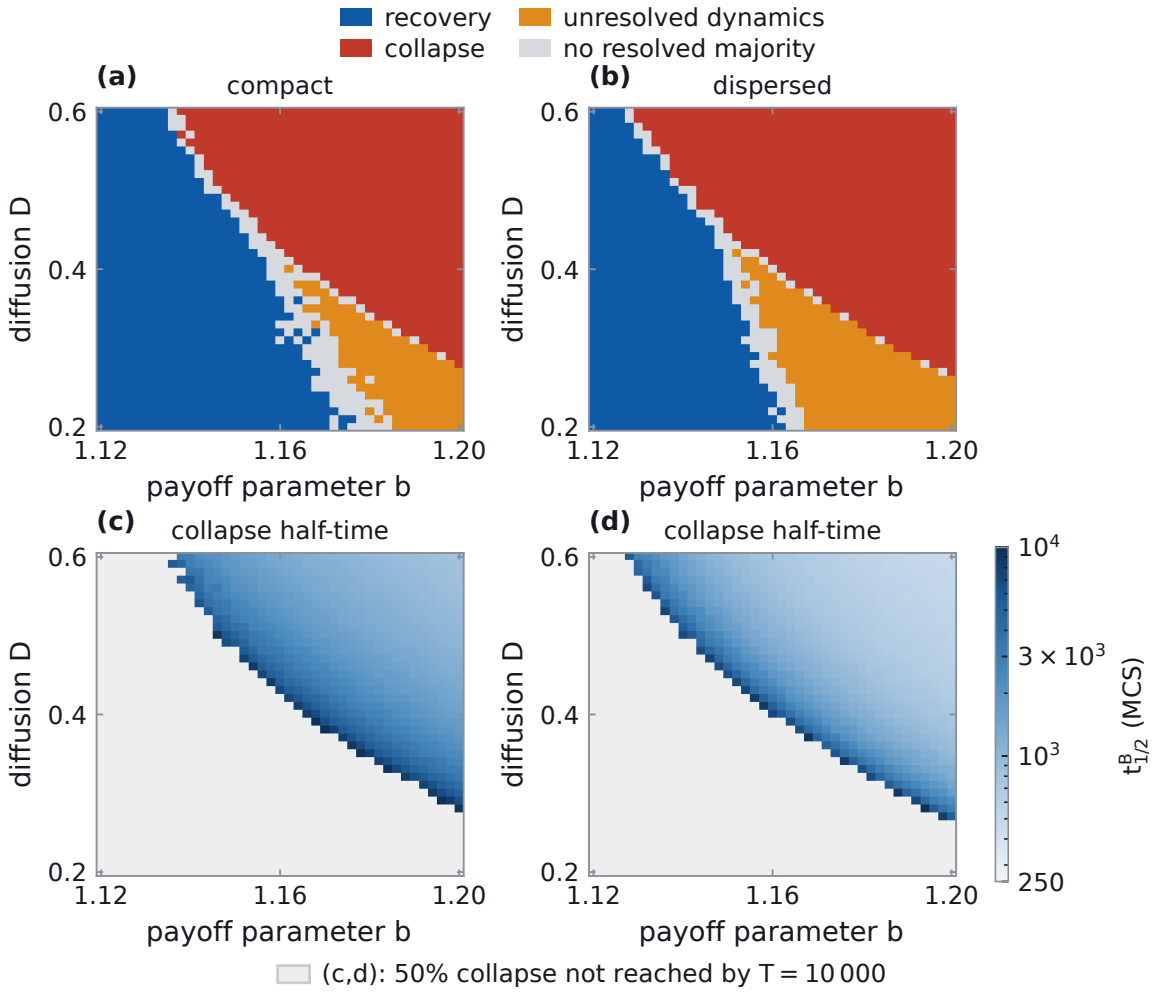


FIG. 1. Finite-time outcomes and collapse times for compact and dispersed perturbations ($L = 48$, $m = 16$, $T = 10\,000$ MCS). (a,b) Parameter regions in which recovery (A), persistent depletion (B), or unresolved evolution (U) has a statistically resolved majority; gray cells have no resolved majority, as defined in Supplemental Material, Sec. SII. (c,d) First time $t^B_{1/2}$ at which 50% of the full 64-trajectory ensemble has recorded event B; gray indicates that this level is not reached within the observation window. Each panel shows the measured 41×41 parameter grid without interpolation. All other parameters take their reference values.

negative for compact ones (Sec. SIII of the Supplemental Material, Fig. S9). The dispersed interface is also substantially longer, producing more opportunities for strategy changes.

Figure 4 follows these effects beyond the first visit. Early in the evolution, the dispersed interface becomes increasingly active even as the normalized signed bias approaches zero. The compact interface remains less active. The upper panels show how these different early responses are reflected in finite-time recovery, collapse, and unresolved probabilities. The normalized quantity $4\langle J_{\mathrm{pre}}\rangle/\langle P\rangle$ is a ratio of ensemble means, not an average of edgewise biases. Equation (8) applies exactly to the first visit conditioned on a fixed interface; it is not an exact formula for the net population change over a complete MCS, during which the strategy configuration evolves. The clipped implementation and its corresponding identities are treated in Sec. SI of the Supplemental Material.

B. Establishment and spatial consolidation

To distinguish establishment from subsequent expansion, let $\mathcal{C}_\alpha$ be the periodic nearest-neighbor components of the defector set. The largest-cluster share is

$$f_{\max}(t) = \frac{\max_\alpha |\mathcal{C}_\alpha(t)|}{n_D(t)}, \qquad n_D = N(1-\rho), \tag{10}$$

with $f_{\max} = 0$ for an empty defector set. Configurations are compared at first crossings,

$$\tau_f = \inf\{t : n_D(t)/N \geq f\}. \tag{11}$$

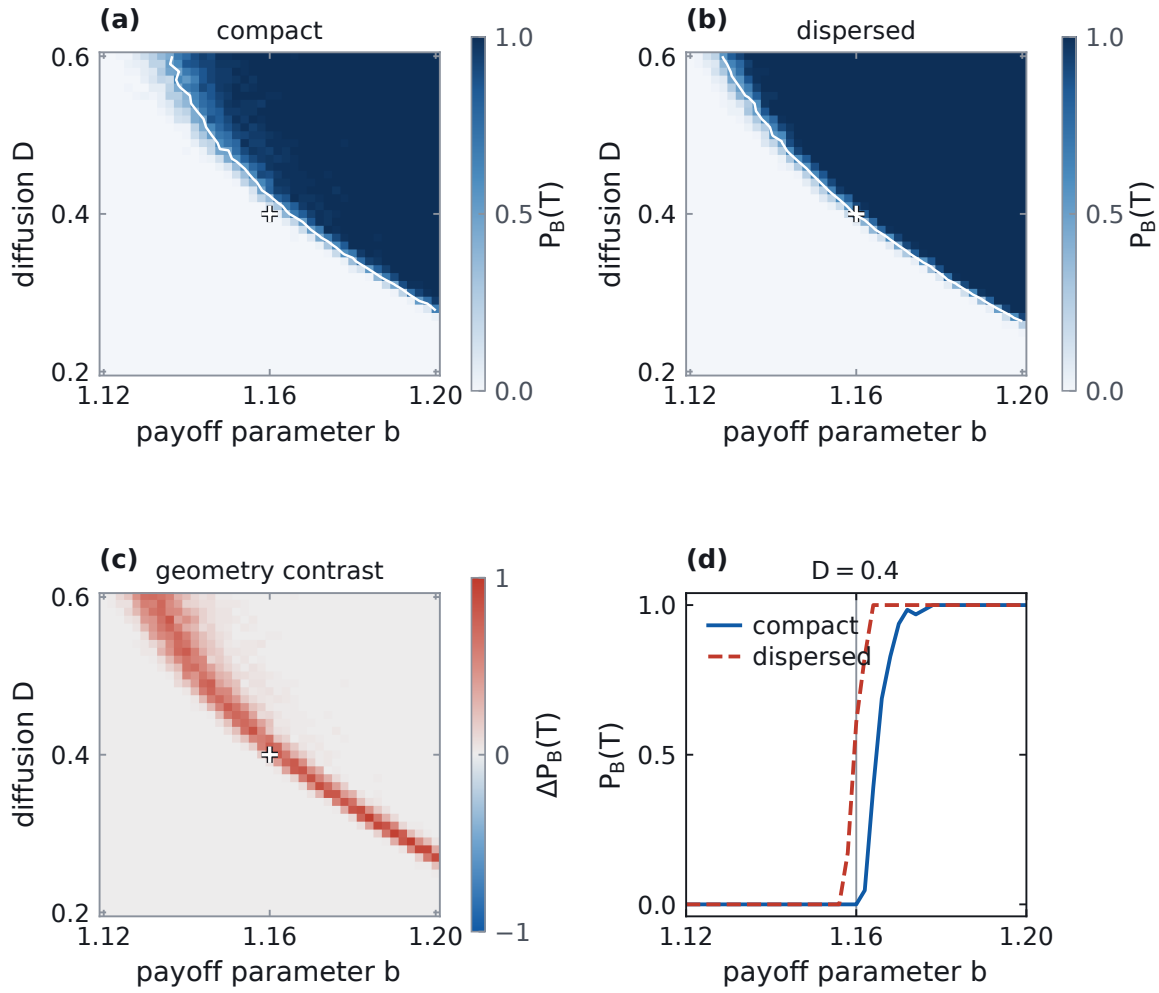


FIG. 2. Collapse-probability contrast between perturbation geometries. (a,b) Finite-time collapse probabilities for compact and dispersed seeds in the (b, D) plane. (c) Difference $P_B^{\text{dispersed}}(T) - P_B^{\text{compact}}(T)$. (d) Cross section at $D = 0.4$. Each grid point represents 64 independent realizations per geometry at $T = 10\,000$ MCS. Cell colors show measured estimates without image interpolation; white half-probability contours are visual guides obtained by interpolation. Crosses indicate the reference parameters, and lines in (d) connect measured values.

The threshold-crossing ensembles support the spatial interpretation in Fig. 3. Near the first 5% crossing, compact-origin disturbances are typically concentrated in one dominant cluster, whereas dispersed-origin disturbances retain several separated clusters. By the first 20% crossing, both preparations are usually dominated by a connected component; the complete cluster-count and arrival-time distributions are shown in Fig. S6.

The geometry contrast is especially pronounced in the probability of reaching the first growth threshold. Because every trajectory that satisfies the collapse criterion must previously have crossed 5% defectors,

$$P_B(T) = \Pr(\tau_{0.05} \le T)\, \Pr(\tau_B \le T \mid \tau_{0.05} \le T). \quad (12)$$

This decomposition separates an establishment factor from a subsequent conditional factor. Dispersed seeds have a much larger establishment probability, whereas the conditional travel times between later population thresholds are more similar. These observations locate an important part of the geometry effect in the early establishment stage, without requiring a constant propagation speed or a universal coarsening law.

The ordering is retained when the initially converted sites are not depleted of resources. Hence the initial resource loss can affect the quantitative outcome but is not required for the observed geometry contrast. What matters is the spatial organization of interacting strategies and the resource field that develops as their dynamics unfold.

V. RESOURCE LAYERS AND RELATIVE FIELD ALIGNMENT

As a disturbance expands, defectors deplete resources in its interior while cooperators maintain higher resource

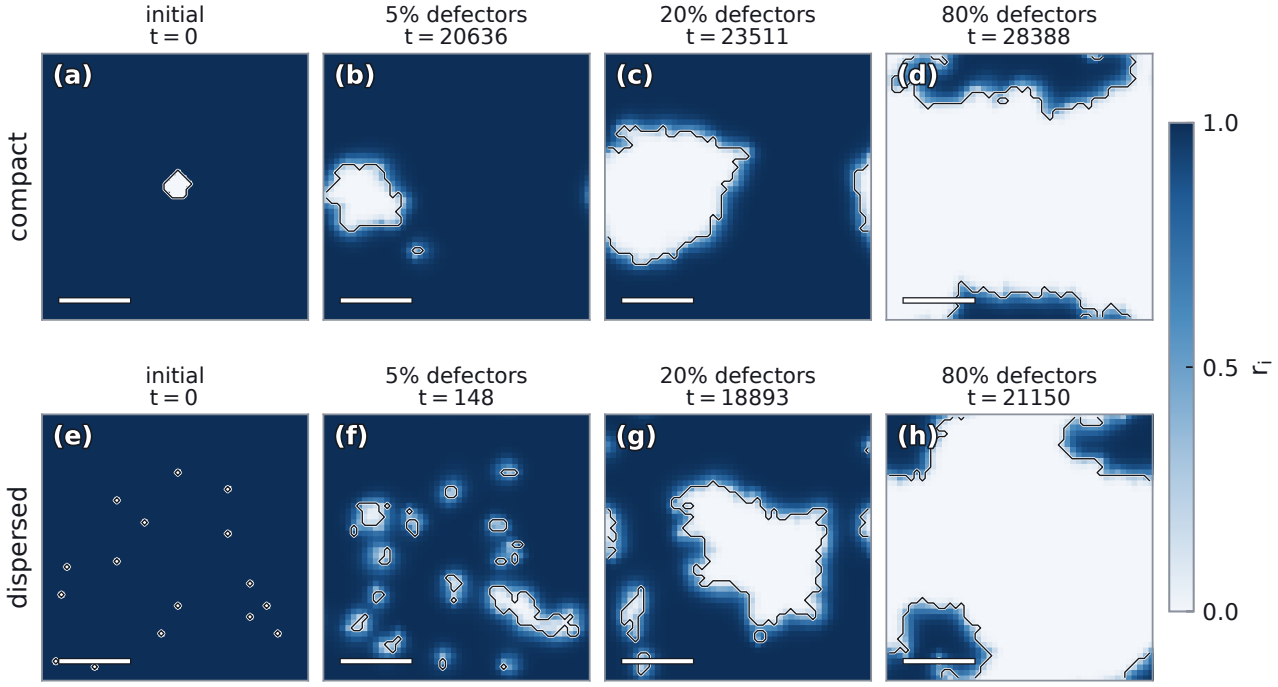


FIG. 3. Spatial growth of finite disturbances and their resource environments. Compact and dispersed perturbations are shown in the upper and lower rows, respectively. Columns show the initial state and the first crossings of 5%, 20%, and 80% defectors; the annotated times are the actual crossing times in MCS and differ between trajectories. Resource colors and strategy-boundary contours distinguish depleted interiors from cooperative surroundings. Both examples were selected from collapse trajectories independently of their appearance. A common resource scale is used throughout; each trajectory is displayed after one fixed spatial translation. Scale bars represent 12 lattice sites. Corresponding strategy maps are shown in Fig. S5.

levels in the surrounding region. Their competition therefore generates a resource gradient across the strategy boundary. To isolate the role of this relative arrangement, we hold the strategy configuration fixed and translate the entire resource field on the periodic lattice,

$$(\mathcal{T}_{\mathbf{d}} r)_i = r_{i-\mathbf{d}}. \tag{13}$$

This operation preserves the total resource, its empirical distribution, and its autocorrelation at every displacement,

$$\Gamma_r(\mathbf{a}) = \frac{1}{N}\sum_i (r_i - \bar{r})(r_{i+\mathbf{a}} - \bar{r}), \tag{14}$$

but generally changes the cross-covariance between the strategy and resource fields,

$$C_{sr}(\mathbf{d}) = \frac{1}{N}\sum_i (s_i - \rho)(r_{i-\mathbf{d}} - \bar{r}). \tag{15}$$

Figure 5 illustrates this controlled comparison. The translated field has exactly the same internal resource structure as the original, but its depleted and resource-rich regions are differently positioned relative to the unchanged strategy interface.

For a grown configuration X in which neither event has occurred, let τ_A and τ_B be the first recovery and collapse confirmation times measured from the continuation origin. The conditional fate probabilities at horizon H are

$$\begin{aligned} q_B(H \mid X) &= \Pr(\tau_B \le H,\ \tau_B < \tau_A \mid X), \\ q_A(H \mid X) &= \Pr(\tau_A \le H,\ \tau_A < \tau_B \mid X), \\ q_U(H \mid X) &= 1 - q_A(H \mid X) - q_B(H \mid X). \end{aligned} \tag{16}$$

We compare these probabilities after a single resource translation of each grown configuration. Both fields then evolve under the original dynamics, retaining the same strategy state at the start of each continuation.

Despite these invariants, the translated configurations have different finite-time collapse probabilities. A short

TABLE I. Translation-induced changes in conditional collapse probability at $H = 40000$ MCS. The comparison uses 48 configurations at their first 5% defector crossing, with 16 fresh continuations per field and direction. Four directions are averaged within each magnitude; 95% bootstrap intervals retain pairing within configurations.

Comparison	Δq_B	95% interval
Short shift–original	0.0498	$[0.0052, 0.0944]$
Long shift–original	−0.0628	$[-0.0983, -0.0273]$
Long–short shift	−0.1126	$[-0.1370, -0.0882]$

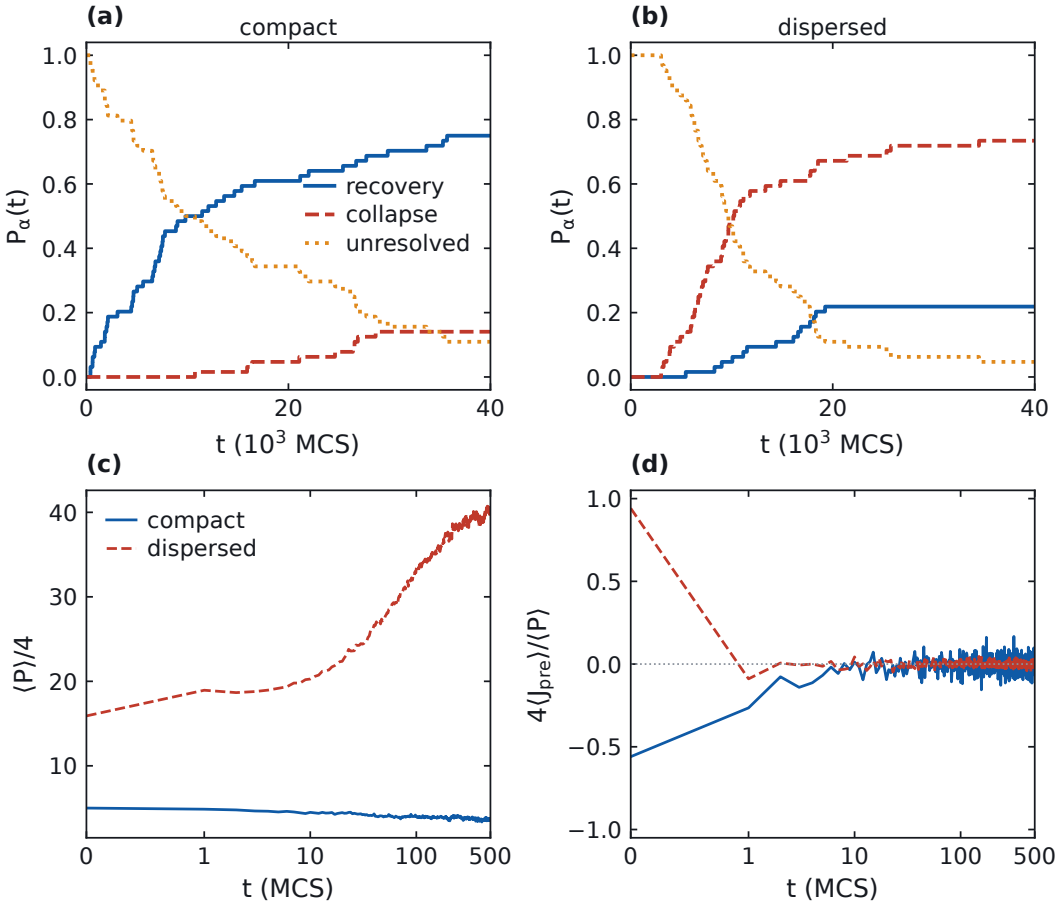


FIG. 4. Interfacial response and finite-time fate. (a,b) Time-dependent recovery, collapse, and unresolved probabilities for compact and dispersed perturbations (64 trajectories per preparation). (c) Mean early interface activity, $\langle P\rangle/4$. (d) Normalized signed bias, $4\langle J_{\rm pre}\rangle/\langle P\rangle$, a ratio of ensemble means that is set to zero when all interfaces vanish. Panels (c,d) use independent 64-trajectory early-time ensembles and retain absorbed trajectories. The lower time axes are linear through 1 MCS and logarithmic thereafter. The upper and lower rows cover distinct time windows and are not joined into continuous trajectories.

translation increases the conditional probability of collapse relative to the original field, whereas a longer translation decreases it. In the paired continuation experiment at $H = 40\,000$ MCS, the changes are $+0.0498$ and -0.0628, respectively (Table I). An independent displacement-length series also shows a turnover rather than a monotonic dependence on translation magnitude. Thus the relative arrangement of the two fields affects the outcome even when the strategy pattern and the single-field resource statistics are held fixed. Displacement vectors, sampling details, and uncertainty estimates are given in Sec. SII of the Supplemental Material.

The interfacial resource profiles in Fig. 6 provide a local view of this effect. Distance is measured relative to the strategy boundary, with negative values inside defector regions and positive values inside cooperative regions. Before translation, the mean resource profile rises sharply across the interface. A short shift broadens this transition, whereas the displayed long shift moves resource-rich regions toward parts of the boundary and raises resource levels at neighboring cooperative sites.

Renewal, depletion, and diffusion subsequently act on the translated field, but a single update does not remove the change in boundary conditions. For the displayed long shift, the interfacial payoff-gap distribution shifts toward values favoring cooperation. Relative field alignment therefore changes the immediate microscopic bias of interface motion. Its effect on collapse probability also depends on how the interface and resource layer evolve after this initial perturbation.

Alignment also affects net resource production through the strategy–resource covariance. Subtracting Eq. (4) for original and translated fields with equal initial strategy and resource means gives

$$\Delta\bar{r}(1) = (\delta - g)\Delta C_{sr}(0). \tag{17}$$

Equation (17) shows that changing relative alignment alters the next mean-resource increment even though the initial resource total is unchanged. Interfacial payoff differences and total resource production thus respond to the same spatial rearrangement through distinct local and global mechanisms. Neither response can be inferred

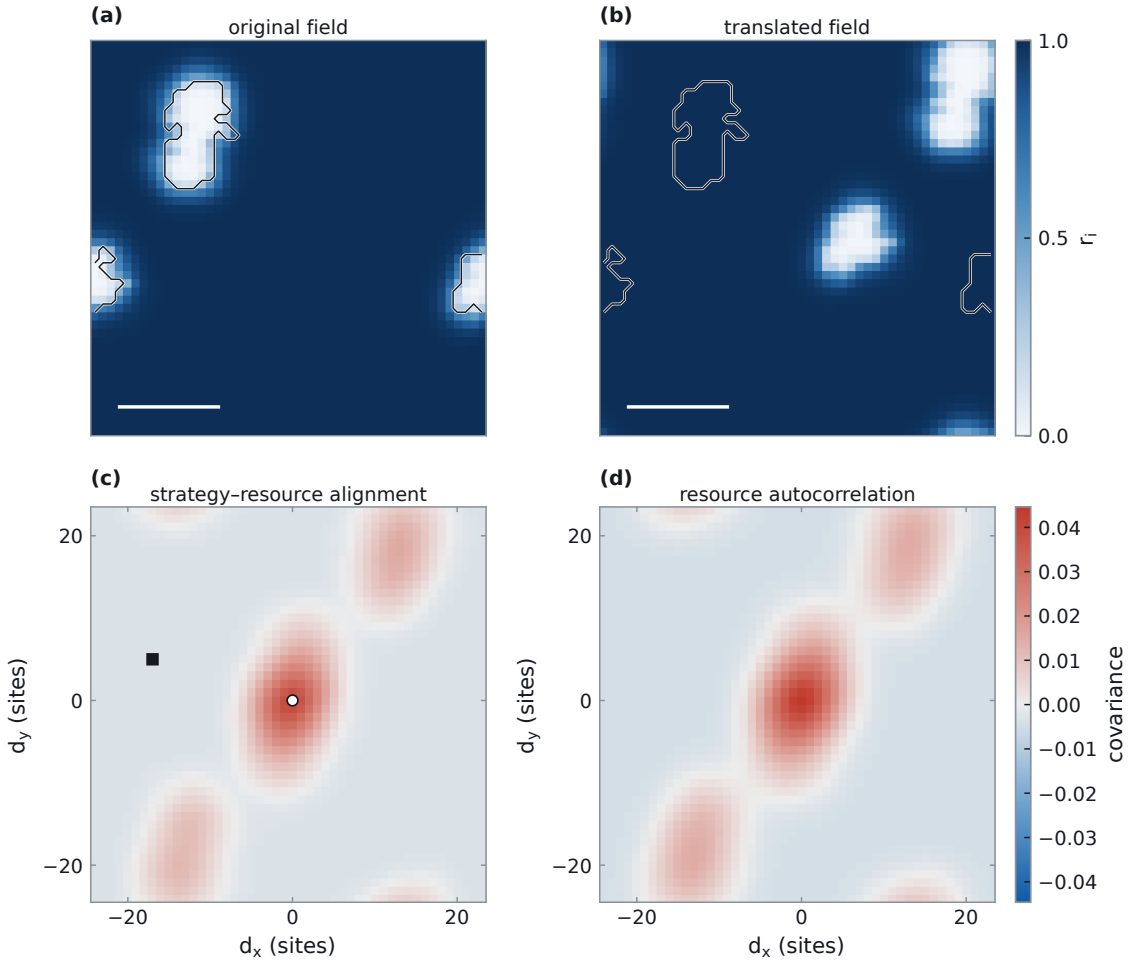


FIG. 5. Controlled change in relative strategy–resource alignment. (a,b) Original resource field and the same field translated by $(5, -17)$ sites, with the strategy boundary unchanged. Scale bars represent 12 sites. (c) Strategy–resource cross-covariance $C_{sr}(\mathbf{d})$ across periodic translations; the circle and square mark the fields in (a,b). (d) Resource autocorrelation $\Gamma_r(\mathbf{d})$, which is invariant under the translation. The covariance maps use a common signed color scale. These panels characterize spatial organization, not collapse probability.

from the initial mean resource alone.

VI. TRANSIENT RESPONSE AND RESOURCE MEMORY

A. Adaptation to a moving strategy pattern

The resource field contains information about the recent spatial arrangement of strategies. Let M_D represent nearest-neighbor diffusion and A_s be diagonal with entries $a_i = 1 - gs_i - \delta(1 - s_i)$. When the strategy configuration is held fixed, resource dynamics take the affine form

$$T_s(r) = M_D(A_s r + gs). \tag{18}$$

For $0 < g, \delta \leq 1$, this map is a maximum-norm contraction under a fixed strategy pattern, with $q = \max\{1 - g, 1 - \delta\} < 1$:

$$\|T_s^t(r) - T_s^t(\widetilde{r})\|_\infty \leq q^t \|r - \widetilde{r}\|_\infty. \tag{19}$$

It therefore has a unique equilibrium resource field $r_{\mathrm{eq}}(s)$. The deviation from that equilibrium can be bounded by the one-step residual,

$$\|r - r_{\mathrm{eq}}\|_\infty \leq \frac{\|r - T_s(r)\|_\infty}{1 - q}. \tag{20}$$

as demonstrated numerically in Fig. S4. This result concerns two resource fields evolving under the same fixed strategy pattern.

In the coupled game, however, strategy updates continually change the equilibrium resource field itself. To describe this mismatch, we measure the mean distance to the instantaneous fixed-strategy equilibrium and compare the actual interfacial bias with its adapted-field

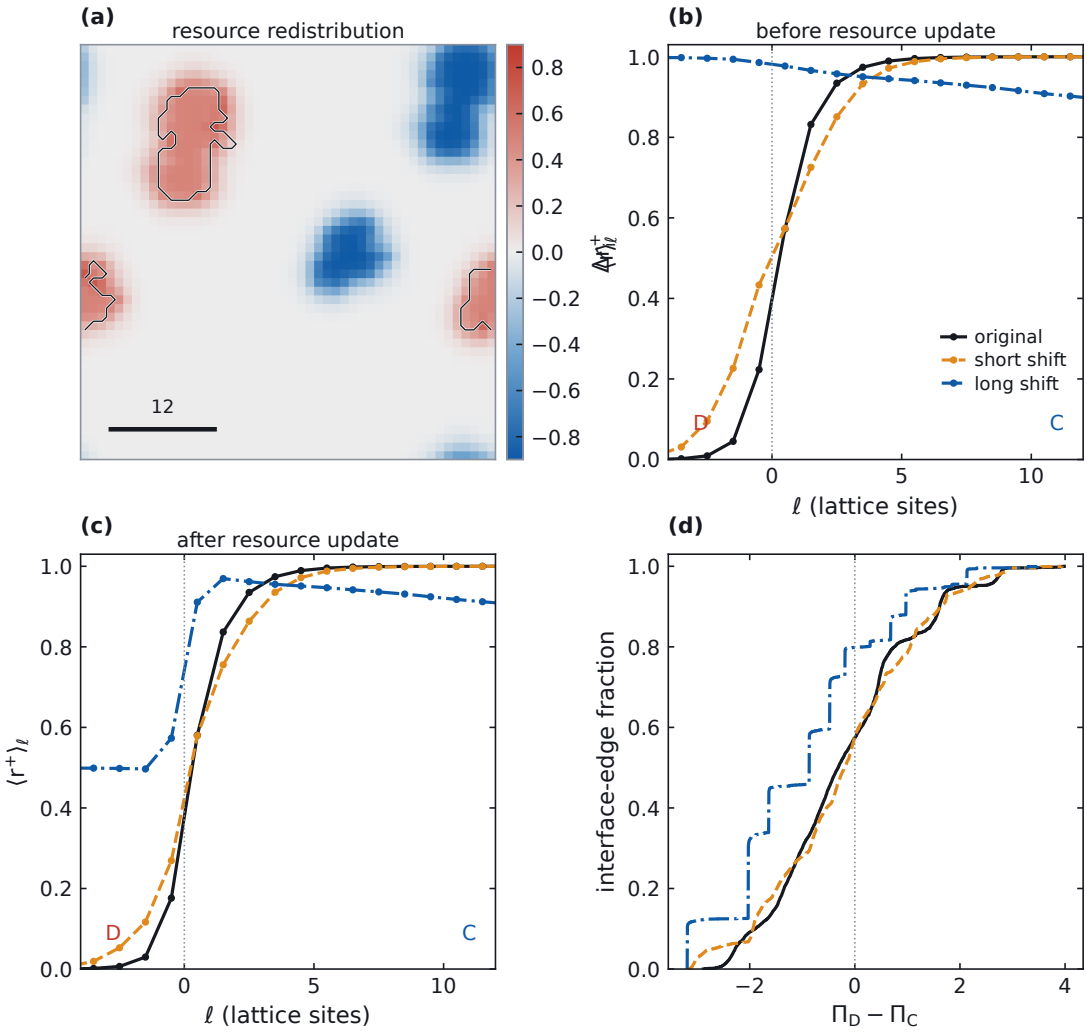


FIG. 6. Resource profiles and payoff gaps near a fixed strategy boundary. (a) Difference in the post-update resource field under the displayed long translation, with the strategy interface unchanged (scale bar: 12 sites). (b,c) Mean resource profiles before and after one resource update, plotted against signed periodic graph distance ℓ from the interface. Negative and positive distances lie in defector and cooperator regions, respectively, with boundary sites at $\ell = \pm 1/2$. (d) Cumulative distribution of interfacial payoff gaps after the resource update. The displayed short and long translations are $(2, 1)$ and $(5, -17)$. Profiles pool lattice sites from 48 configurations within each distance shell; gap distributions pool undirected cooperator–defector edges. Only the displayed displacement vectors are used in these panels, without rotation averaging.

counterpart:

$$\eta(s,r) = \frac{1}{N}\sum_i |r_i - r_{\mathrm{eq},i}(s)|, \qquad J_{\mathrm{eq}}(s) = J(s, r_{\mathrm{eq}}(s)). \tag{21}$$

The mismatch η measures incomplete resource adaptation, while $J - J_{\mathrm{eq}}$ compares the current interfacial bias with that obtained from a resource field equilibrated to the same strategy configuration. These diagnostics distinguish relaxation toward a fixed target from evolution in which the target moves. Consequently, the contraction bound for fixed strategies does not define a relaxation time for the fully coupled dynamics.

B. Spatial sensitivity and the loss of initial ordering

Different parts of the interface have different sensitivities to resource changes. On the positive reward branch, differentiating the signed bias with respect to the resource level of a boundary cooperator yields

$$w_u = -\frac{\partial J}{\partial r_u} = \frac{\lambda\sigma k_C(u)}{4K}\sum_{v\in\mathcal{N}_D(u)}[1-\tanh^2 z_{uv}], \qquad z_{uv} = \frac{\Pi_D(v) - \Pi_C(u)}{2K}. \tag{22}$$

Only boundary cooperators contribute directly to this derivative. Resource perturbations introduced before the resource substep are additionally redistributed by local

reaction and diffusion, giving the effective first-order response

$$\delta J_{\mathrm{pre}} = -\sum_u a_u W_u h_u + O(\|h\|^2),$$
$$W_u = (1-D)w_u + \frac{D}{4}\sum_{v\sim u} w_v, \tag{23}$$

The sensitivity weights are evaluated at the unperturbed post-update field. The remainder is quadratic for a fixed lattice and parameter set, provided the perturbation remains on the positive reward branch. For boundary cooperators, the local reaction factor is $a_u = 1 - g$.

Figure 7 visualizes these spatially heterogeneous sensitivities. Assigning the largest existing boundary resources to the largest effective weights minimizes the linearized signed growth bias; reversing the assignment maximizes it. Both rearrangements preserve the multiset of boundary resource values, total resource, and initial strategy–resource covariance. Because the rearrangement argument concerns a linearized objective, the finite changes in J_{pre} are evaluated using the full nonlinear expression rather than assumed to attain the same extrema.

The two arrangements produce contrasting initial responses, but this ordering does not determine later population trajectories. To separate immediate drift from accumulated changes, define $a_a(t) = \mathbb{E}_a[n_D(t+1) - n_D(t)]$, with $a = 0$ denoting the original resource field. Because all treatments begin with the same strategy configuration, the cumulative difference satisfies

$$\mathcal{I}_a(t) = \mathbb{E}_a[n_D(t)] - \mathbb{E}_0[n_D(t)] = \sum_{u=0}^{t-1}\big[a_a(u) - a_0(u)\big]. \tag{24}$$

Figure S10 shows that a resource assignment that initially suppresses defector growth can produce a positive accumulated population difference at later measured times. Thus a first-step response does not establish a persistent ordering of population changes. The longer-time continuation experiment likewise does not statistically resolve an ordering of collapse probabilities between the two local assignments; this should not be interpreted as evidence that their outcomes are identical.

Resource translations and boundary permutations probe different features of spatial organization. Because their initial signed biases are not matched, they do not constitute equal-strength interventions. The comparisons establish differences under the interventions actually applied, not a universal ranking of their persistence. Numerical results and uncertainty estimates appear in Sec. SII of the Supplemental Material.

C. Coupled evolution of the two fields

The exact resource balance helps distinguish the two interventions. Translating the resource field immediately changes the strategy–resource covariance, which can alter the next mean-resource increment through Eq. (17). In contrast, permuting resource values among boundary cooperators preserves the strategy–resource covariance, mean resource, and mean strategy. It therefore leaves the next total-resource increment unchanged while potentially redistributing payoff bias along the interface. Subsequent strategy changes create new covariance differences and corresponding changes in resource production.

Figure S11 displays these distinct transient responses through mean population–resource paths and reconstructed covariance changes. Translation changes how the resource landscape is positioned relative to the strategies. Boundary permutation instead changes where resources act along an unchanged interface while preserving the global initial covariance. Their subsequent trajectories reflect the coupled evolution of strategies and resources. The plotted ensemble means are descriptive and do not define a closed deterministic flow in $(\rho, \bar{r})$.

The covariance term in the exact balance applies separately to each trajectory before ensemble averaging. Together with the spatially varying interface sensitivity, it explains why mean population and resource alone omit information relevant to the disturbance response. Sec. SI of the Supplemental Material gives the derivations.

VII. DISCUSSION

Finite disturbances in this spatial feedback game are governed by two related but distinct aspects of spatial organization. First, the initial arrangement of defectors determines the cooperator–defector contacts available for strategy updates. Dispersed seeds have a longer, initially expansion-biased interface and are more likely to reach the early growth threshold than compact seeds with the same size and resource deficit. Second, as a disturbance grows, the evolving strategies create a depleted interior and an interfacial resource layer. The position of that layer relative to the strategy boundary affects the probability of subsequent collapse. Resource translations separate this alignment effect from changes in strategy geometry or in the resource field's own distribution and autocorrelation.

This second result identifies a limitation of descriptions based on separate statistics of the two fields. Network reciprocity explains how the arrangement of strategic interactions affects selection [17, 19], while local environmental feedback makes the payoffs themselves depend on resources generated or depleted by the competing populations [28, 29]. In the present model, states with identical strategy configurations and identical single-field resource statistics can nevertheless have different finite-time collapse probabilities. The strategy–resource covariance and the distribution of local interfacial payoff gaps provide complementary descriptions of this relative alignment, although neither has been shown to determine

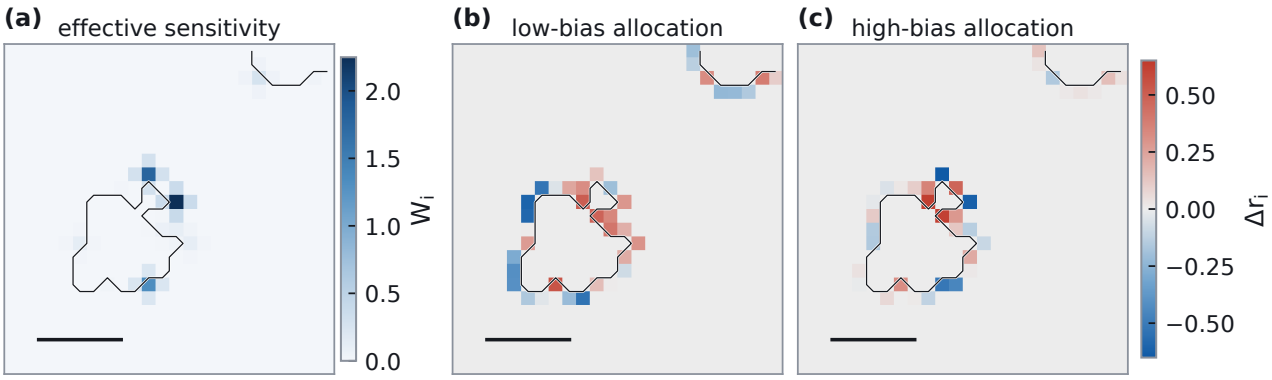


FIG. 7. Spatially heterogeneous interface sensitivity and resource rearrangements. (a) Effective sensitivity weights W_i. (b,c) Resource changes under the low-bias and high-bias assignments of the same existing boundary resource values. Each panel shows an unchanged strategy interface in a common 24×24 periodic window (scale bar: six sites). The signed resource increments share a common color scale. Both assignments preserve the resource total, its empirical distribution, and the initial strategy–resource covariance.

collapse probability on its own.

The exact relations clarify different parts of the mechanism. The first-visit identity separates the frequency of strategy-changing events from their directional bias. The resource-balance identity shows how strategy–resource covariance contributes to the next mean-resource increment. The sensitivity calculation identifies where a resource perturbation most strongly affects the signed interface bias. These statements are exact for their specified updates and assumptions; the measured collapse probabilities reflect subsequent stochastic evolution. In particular, reversing the first-step bias by rearranging boundary resources does not guarantee a corresponding reversal in later population trajectories or in the finite-time collapse probability.

These observations connect the game to the physics of interfaces evolving through a self-modified medium. Theories of front propagation describe how local rules govern moving boundaries [7], and spatial ecological models show how environmental feedback interacts with domain growth and collapse [9, 10]. Here the resource field relaxes toward a configuration-dependent target, but strategy changes continually shift that target. The breakdown of a simple ordering between initial interface response and later dynamics shows why a description based only on instantaneous bias is incomplete. A reduced interfacial theory would need to retain the relevant environmental response or justify its elimination through a controlled approximation.

The scope of these conclusions is finite-time and model-specific. The formation of a dominant defector cluster does not establish a universal coarsening mechanism; distinguishing curvature-driven motion from other sources of interface evolution would require additional length-scale and size-dependent measurements [38, 39]. Likewise, the observed collapse crossovers are defined by first-passage criteria rather than by thermodynamic phase boundaries. The perturbations are externally imposed on an absorbing cooperative background, so the present experiments do not identify a spontaneous nucleation rate or a critical droplet of the kind considered in metastable decay and stochastic escape theory [40, 41]. Some resource-alignment contrasts also weaken or become statistically unresolved under changes in system size or depletion threshold, as reported in Supplemental Material, Sec. SII.

The principal finding is therefore not a universal collapse threshold, but a controlled separation of two contributors to finite-disturbance response: the geometry of the initial strategy interface and its alignment with a dynamically generated resource field. Further comparisons at matched interface activity and initial bias, together with systematic variation of disturbance and system sizes, could test which aspects of this coupled spatial history persist beyond the parameter regime studied here.

DATA AVAILABILITY

The simulation code, the analysis and plotting sources, and the archived trajectory data supporting this article are available in the Stochastic-Nucleation-Game repository at `https://github.com/BKZhao/Stochastic-Nucleation-Game.git`. The archiving protocol and checksum records for the trajectory data, together with the current release status of the newer parameter-grid and continuation data, are described in Supplemental Material, Sec. SII.

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